

Chapter 3-2

Digital Transmission

Topics

1. Digital to digital conversion
2. Analogue-to digital conversion
3. Transmission mode

Digital Transmission

- Transmission of digital signal through the channel is called digital transmission.
- Data
 - Digital
 - Analog
- Signals
 - Digital
 - Analog

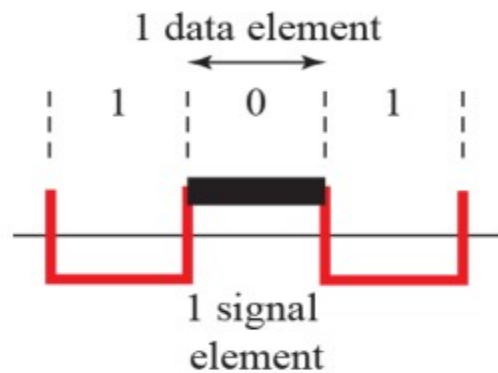
Digital Transmission

- **Digital data , Digital Signals**
 - The simplest form of encoding of digital data is to assign one voltage level to binary one and another voltage level to the binary zero.
 - More complex encoding schemes are used to improve performance by altering the spectrum of the signal and providing synchronization capability
- **Analog data, digital signals**
 - Analog data such as voice and video are often digitized to be able to use digital transmission facilities. The simplest technique is PCM which involves sampling analogue data periodically and quantizing the sample.

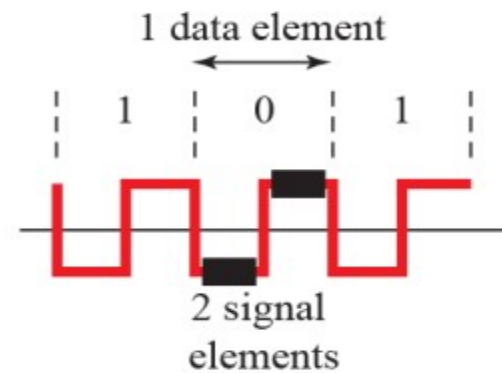
DIGITAL-TO-DIGITAL CONVERSION

- Some characteristics
 - Signal Element V.s Data Element
 - Data Rate V.s Signal Rate
 - Bandwidth
 - Baseline Wandering
 - DC Components
 - Self-synchronization
 - Built-in Error Detection
 - Complexity

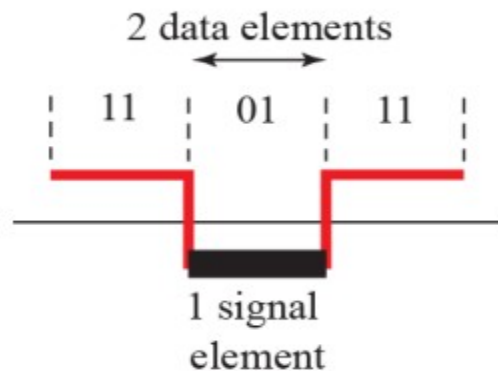
term	Unit	Definition
Data element	bit	A single (binary one or zero) entity that can represent a piece of information
Data rate /Bit Rate	Bits per second (bps)	The number of data elements sent in 1s.
Signal element	Digital : a voltage pulse of constant amplitude Analogue: a pulse of constant frequency phase and amplitude	That part of a signal that occupies the shortest interval of the signaling code. The shortest unit (time wise) of a digital signal
Signaling rate/modulation rate /pulse rate/ baud rate	Signal element per second (baud)	the number if signal elements sent in is.



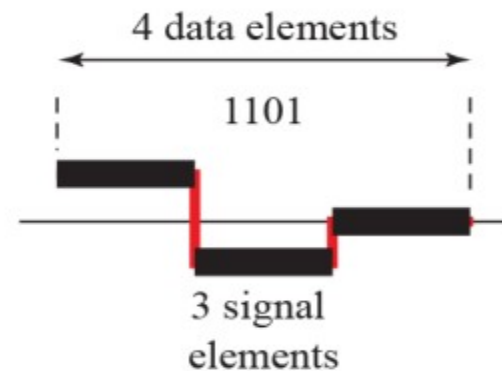
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

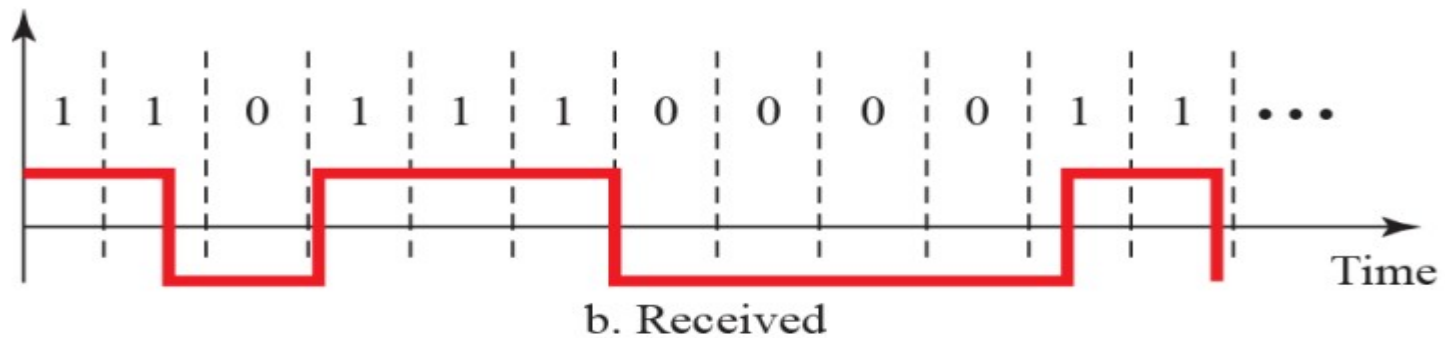
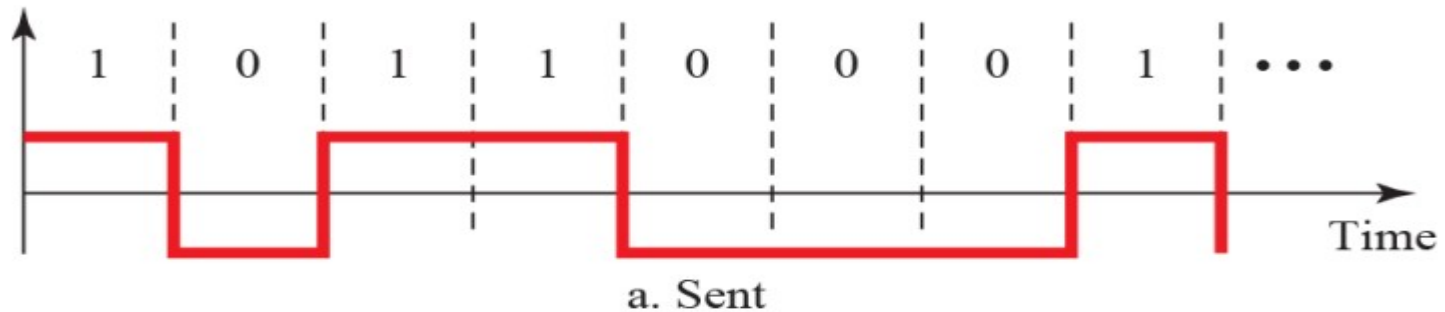
- **DC component**

- Residual Direct Current(DC) components or zero frequencies are undesirable
- Some systems do not allow passage of DC component (such as a transformer) may distort the signal and create output errors
- Dc components is extra energy residing on the line and is useless

- **Self-synchronization**

- Digital signal includes timing information in the data being transmitted to prevent miss interpretation
- Receiver setting the clock matching the sender's

Effect of lack of synchronization



DIGITAL-TO-DIGITAL CONVERSION

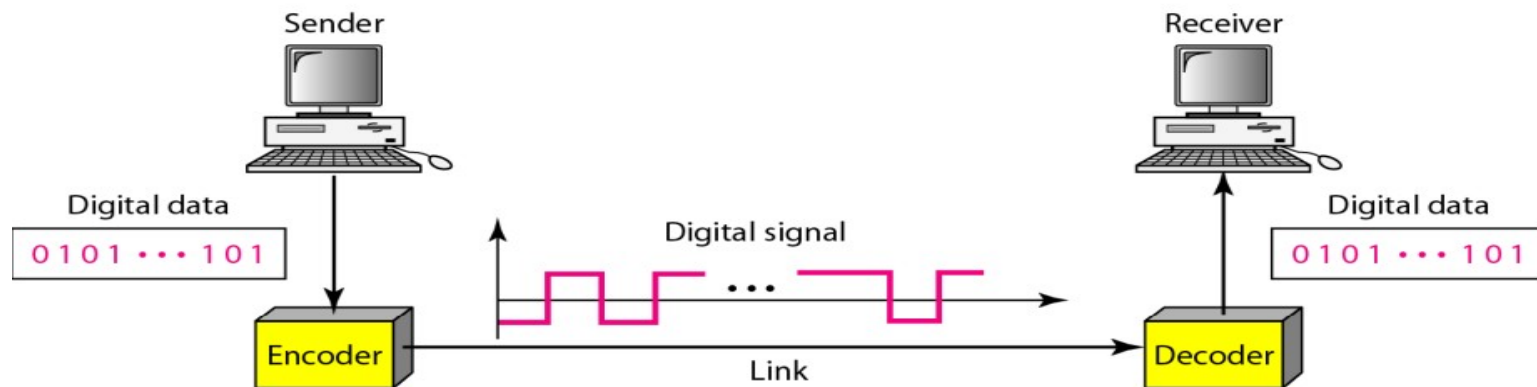
- Three techniques to convert digital data to digital signal.
 - **line coding**
 - **block coding**
 - **scrambling**
- Line coding is always needed; block coding and scrambling may or may not be needed

- three techniques to convert digital data to digital signal.
 - **line coding**
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- Line coding is always needed; block coding and scrambling may or may not be needed.

Line coding

- Line coding is the process of converting digital data to digital signal

Line coding and decoding

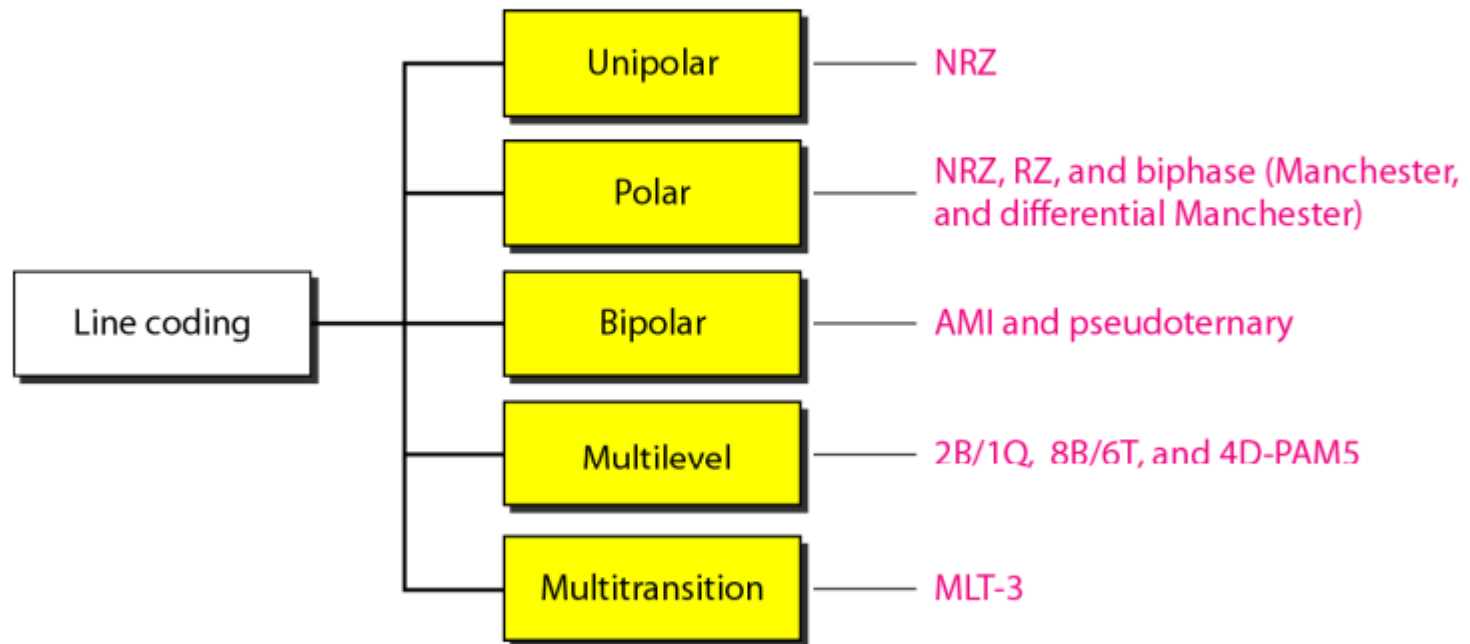


- One goal in data communications is to **increase the data rate** while **decreasing the signal rate**.
- Increasing the data rate increases the speed of transmission; decreasing the signal rate decreases the bandwidth requirement.

Note

Although the actual bandwidth of a digital signal is infinite, the effective bandwidth is finite.

Line coding schemes

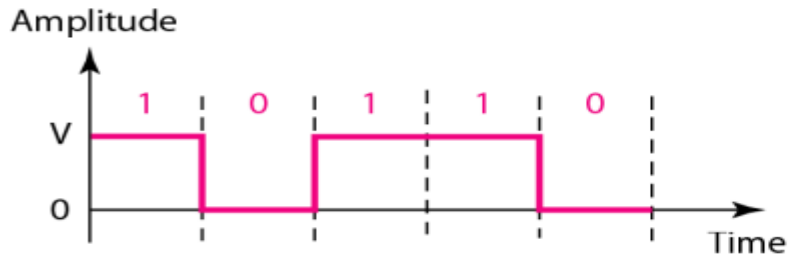


Unipolar

- Simplest method ; inexpensive
- Uses only one voltage level
- Polarity(+ or-) is usually assigned to binary 1 and a 0 is represented by zero voltage.
- Potential problems
 - DC component
 - Lack of synchronization

Unipolar Non Return to Zero(NRZ)

Unipolar NRZ scheme



$$\frac{1}{2}V^2 + \frac{1}{2}(0)^2 = \frac{1}{2}V^2$$

Normalized power

Unipolar - All signal elements have the same sign
It is called NRZ because the signal does not return to zero at the middle of the bit.

Polar

- Uses two voltage levels, one positive and one negative
- Alleviates DC component

Non Return to Zero (Polar NRZ)

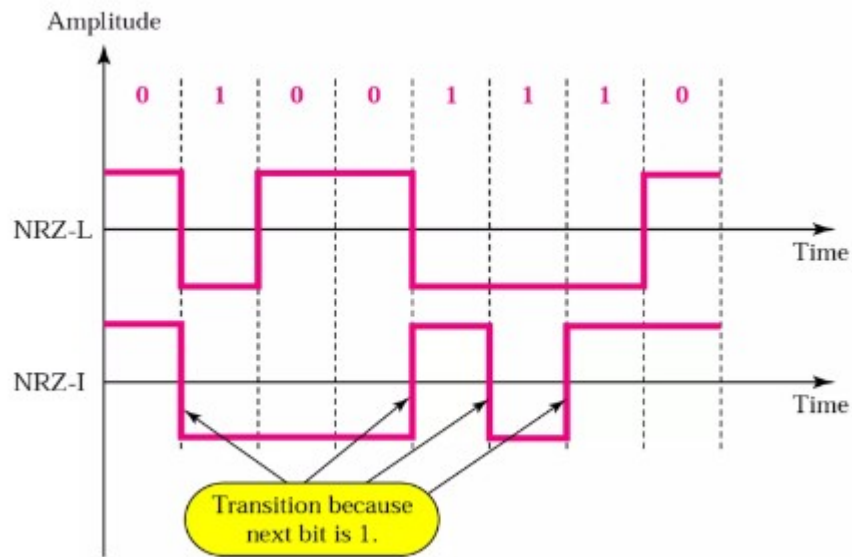
- It uses two different voltage levels(positive or negative) to represent binary values.
- It is also NRZ because there is no rest condition.
- NRZ scheme has two variants:
 - **NRZ-L(NRZ-Level)**
 - **NRZ-I(NRZ-Inverted)**

NRZ-L(NRZ-Level)

- Sign level depends on bit represented
 - 0- high level(positive)
 - 1- low level(negative)
- voltage is constant during bit interval
- Problem : synchronization of long streams of 0s and 1s.

NRZ-I(NRZ-Inverted)

- Inversion of voltage represents 1 → transition (low to high or high to low) at the beginning of the interval.
- 0- no transition at the beginning of the interval (one bit time)
- constant voltage pulse for duration of bit
- Allows for synchronization.



NRZ-L changes voltage level when a different bit is encountered whereas NRZ-I changes voltage when a 1 is encountered.

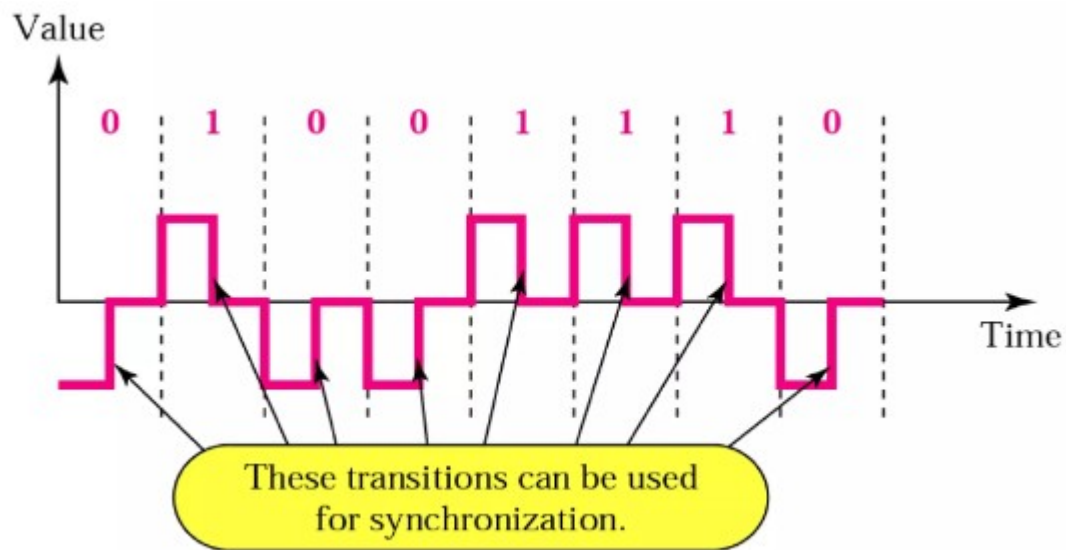
Note

**In NRZ-L the level of the voltage
determines the value of the bit.**

**In NRZ-I the inversion
or the lack of inversion
determines the value of the bit.**

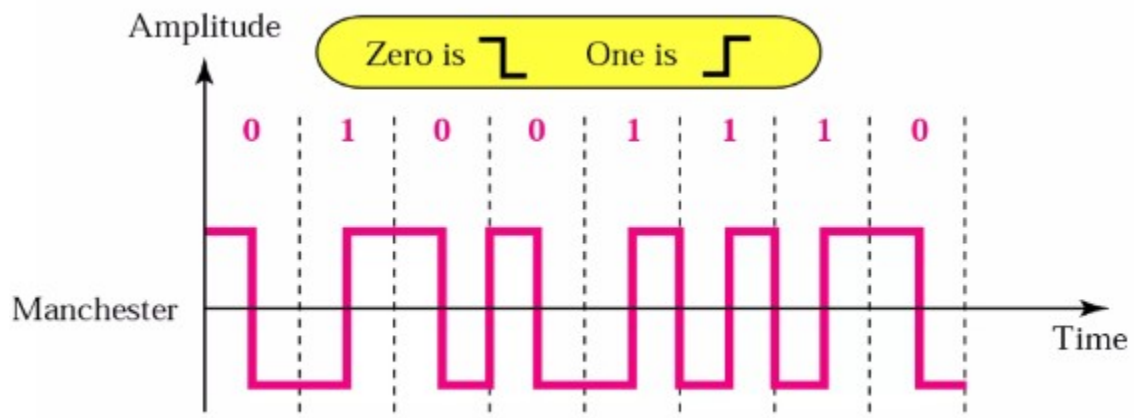
Return to Zero(RZ)

- uses three voltage levels: positive, negative and zero
 - 1 bit is represented by positive-to-zero
 - 0 bit is represented by negative -to-zero
 - zero voltage for none.
- Signals change during bits not between bits.



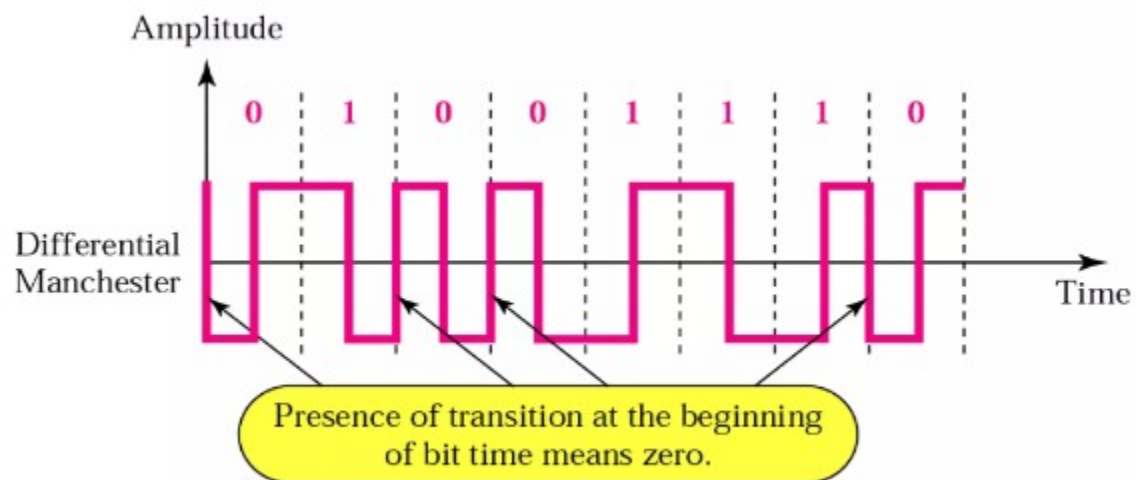
Manchester

- This encoding scheme is a combination of RZ and NRZ-L.
- Uses an inversion at the middle of each bit interval for both synchronization and bit representation(transition serves as clock and data
- Bit time is divided into two halves. It transits in the middle of the bit and changes phase when a **different** bit is encountered.
- 1- transition from **low to high(negative-to-positive)** in the middle of the interval
- 0- transition from **high to low (positive –to negative)** in the middle of the interval



Differential Manchester

- This encoding scheme is a combination of RZ and NRZ-I.
- Always a transition in the middle of the bit interval but changes phase only when 1 is encountered.
- Mid bit transition(inversion) is used clocking (synchronization) only
- transition at start of bit period representing 0
- no transition at start of bit period representing 1
- Requires two signal changes to represent binary 0 but only one to represent 1.



Note

In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for synchronization.

Note

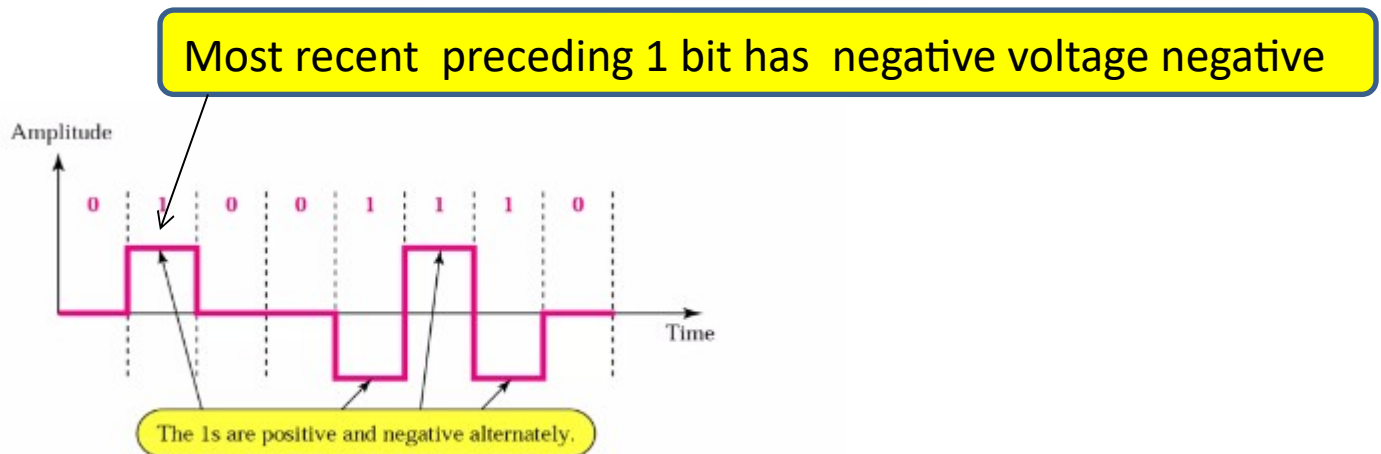
The minimum bandwidth of Manchester and differential Manchester is 2 times that of NRZ.

Bipolar Encoding

- Sometimes called multilevel binary
- Uses three voltage levels: positive, negative, and zero.
- The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative.
- Zero level represents binary 0; 1s are represented with alternating positive and negative voltages even when the 1 bits are not consecutive.
- Two variations of bipolar encoding
 - AMI(alternate mark inversion)
 - Pseudoternary

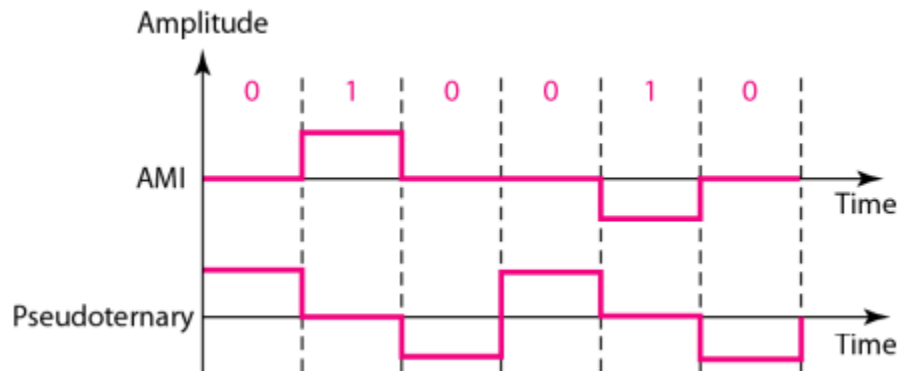
Bipolar-alternate mark inversion (AMI)

- The word mark comes from the telegraphy and means 1
- AMI means alternate 1 inversion
- 0- no line signal (neutral zero voltage)
- 1- represented by positive or negative level voltages alternating for successive ones.



Multilevel Binary Pseudoternary

- Same as AMI but
 - 1- no line signal(neutral zero voltage)
 - 0- alternating positive and negative level(voltage), alternating for successive zeros



Other schemes

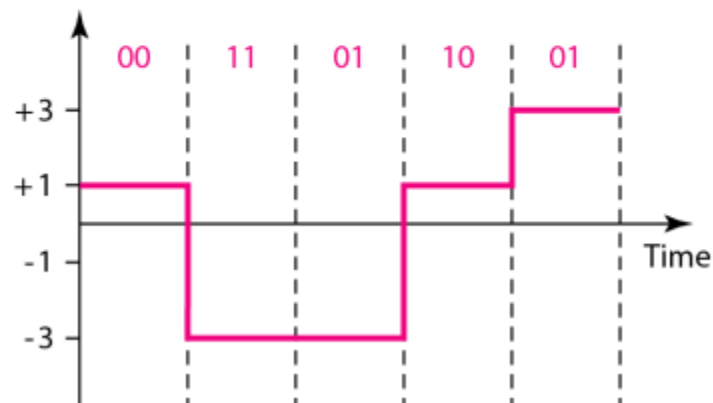
- **2B1Q**(two binary, one quaternary) uses four voltage levels
 - One pulse can represent 2 bits; more efficient.

Multilevel: 2B1Q scheme

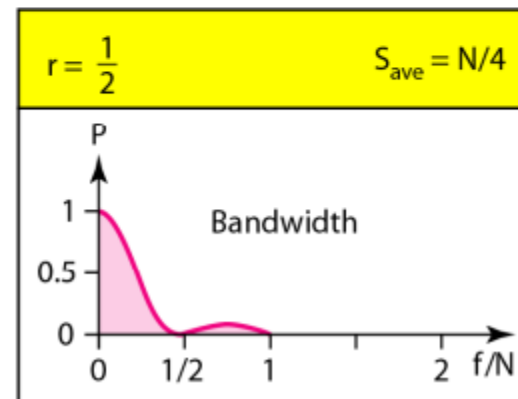
Previous level: positive Previous level: negative

Next bits	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

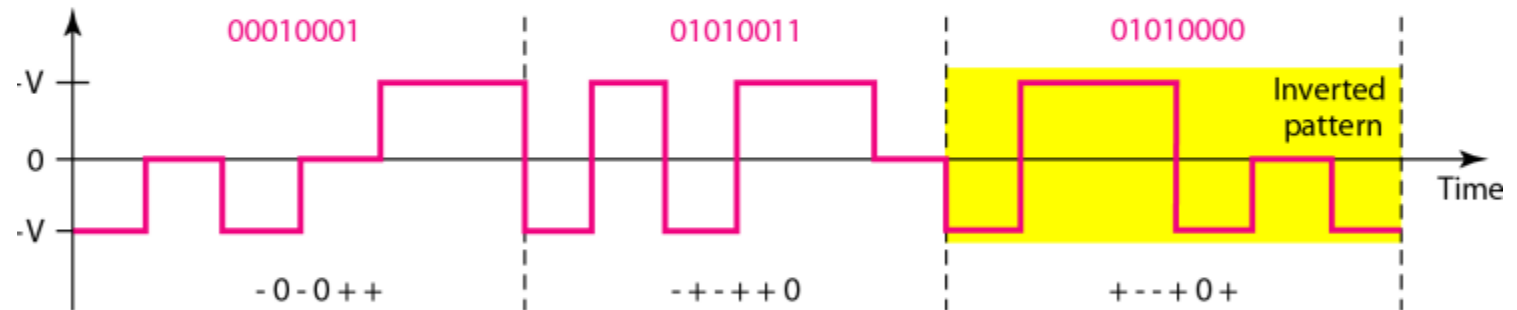
Transition table



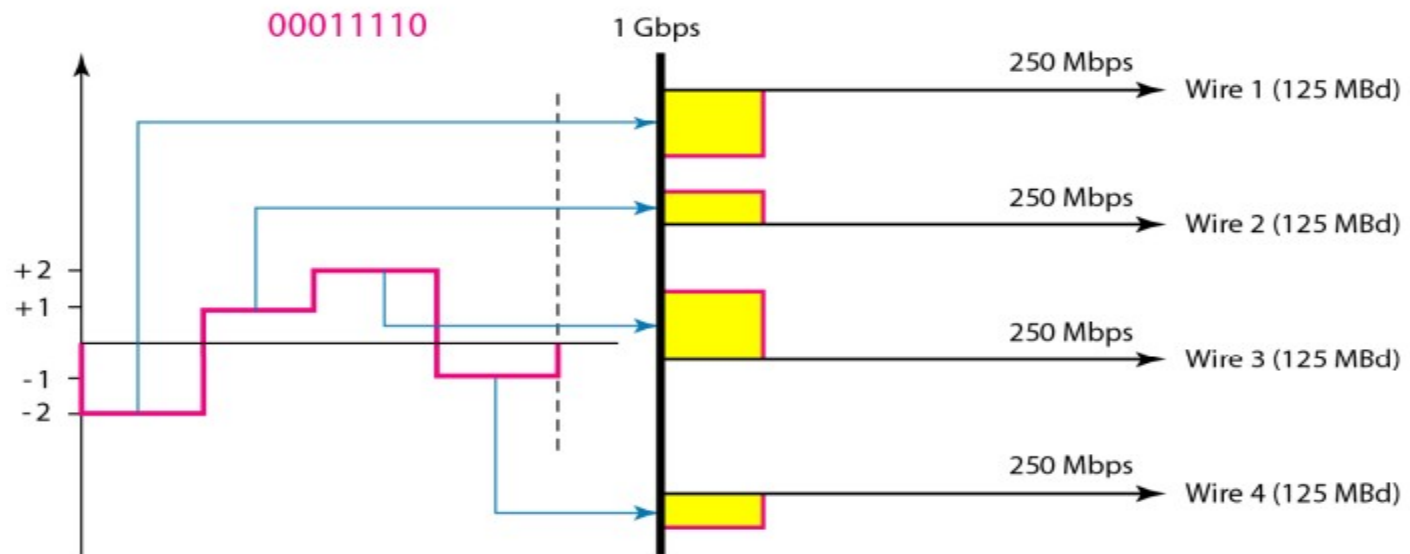
Assuming positive original level



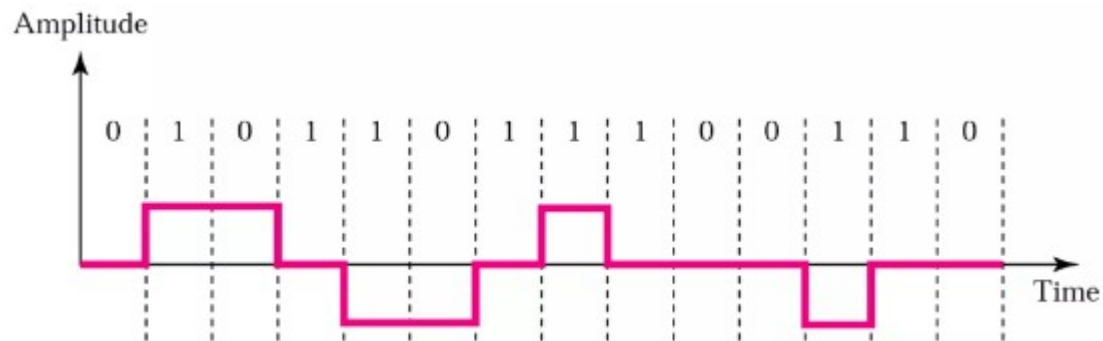
Multilevel: 8B6T scheme



Multilevel: 4D-PAM5 scheme



- **MLT-3**(multi-line transmission, three level)- similar to NRZ-1 using three levels of signals; signal transitions occur at the beginning of 1 bit, no transition at the beginning of 0.



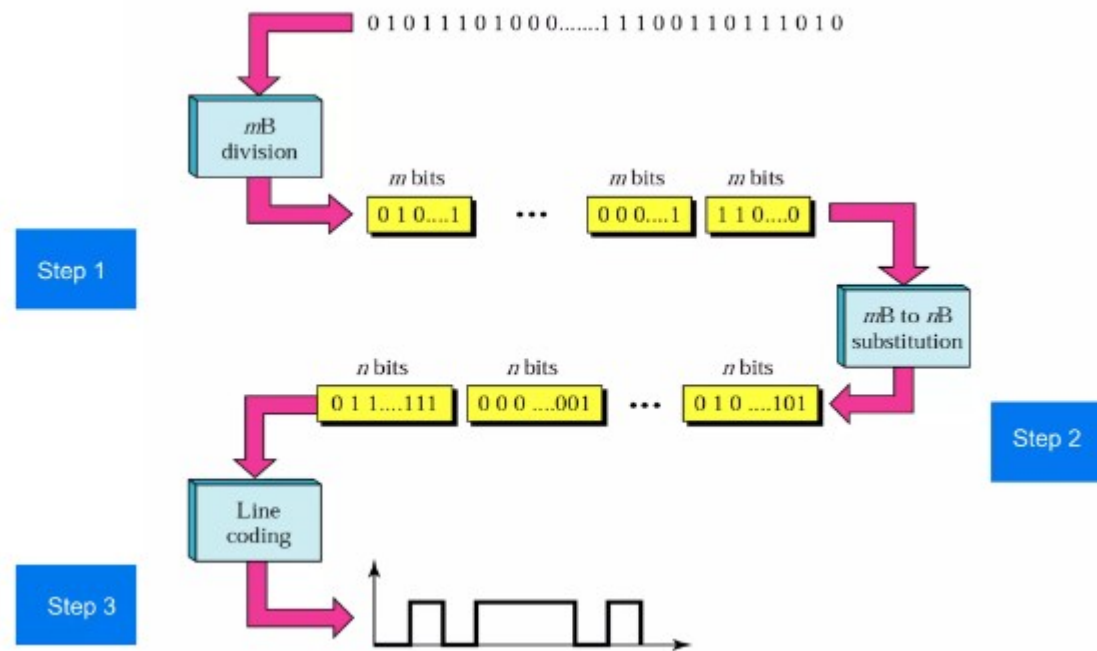
Summary of line coding schemes

<i>Category</i>	<i>Scheme</i>	<i>Bandwidth (average)</i>	<i>Characteristics</i>
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Unipolar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multiline	MLT-3	$B = N/3$	No self-synchronization for long 0s

Block Coding

- Coding method to ensure synchronization and detection of errors.
- Block coding is represented by slash notation, mB/nB . Means, m -bit block is substituted with n -bit block where $n > m$.
- Block coding involves three steps:
 - Division
 - Substitution
 - Combination.
- After block coding is done, it is **line coded** for transmission.

Block coding concept



Transformation Steps

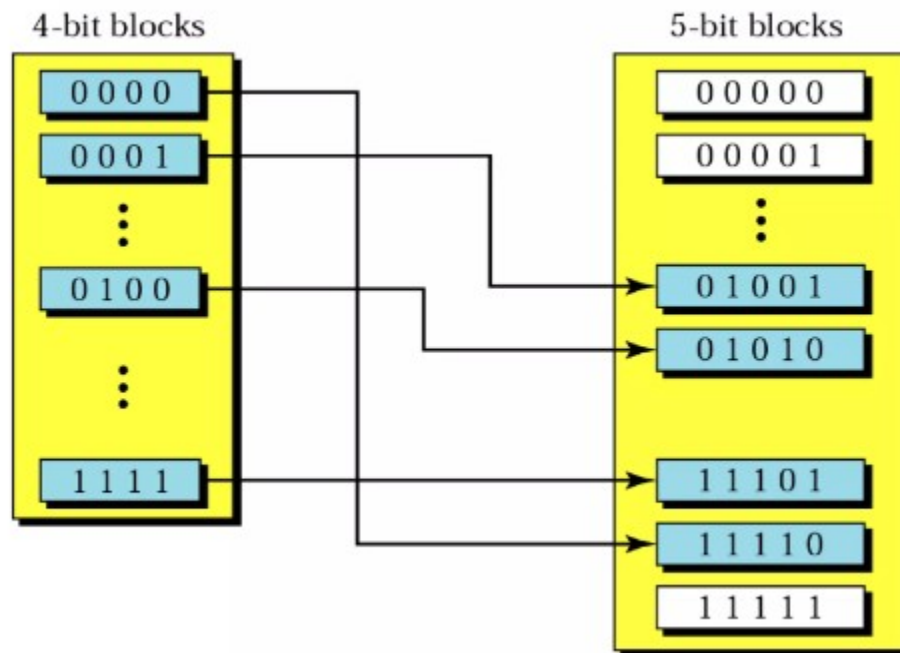
Step 1: bit stream is divided in to groups of m bits

Step 2: substitute an m -bit code for an n -bit group

- Codes with no more than three consecutive 0's or 1's are used to achieve synchronization
- Since only a subset of blocks are used, if one or more bits are changed and invalid code is received a receiver can easily detect the error

Step 3: line encoding scheme is the used to create a signal

- Common block codes
 - **4B/5B**- every 4 bit of data is encoded into a 5 bit code; NRZ-1 is usually used for line coding
 - **8B/10B**- group of 8 bits of data is substituted by a 10 bit code
 - **8B/6T**- each 8 bit group is substituted with six symbol code; uses less bandwidth since three levels may be used



Analog-to-Digital Conversion

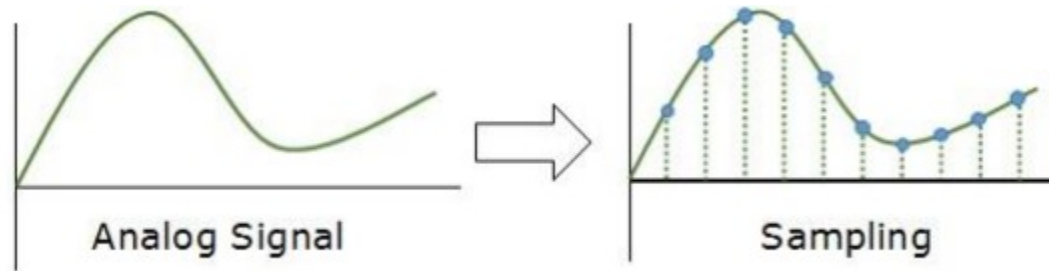
- Microphones create analog voice and camera creates analog videos, which are treated as analog data.
- To transmit this analog data over digital signals, we need **analog to digital conversion**.
- Analog data is a continuous stream of data in the wave form whereas digital data is discrete.
- To convert analog wave into digital data, we use **Pulse Code Modulation (PCM)**.

Pulse Code Modulation (PCM)

- PCM is one of the most commonly used method to convert analog data into digital form.
- It involves three steps:
 - Sampling
 - Quantization
 - Encoding.

Sampling

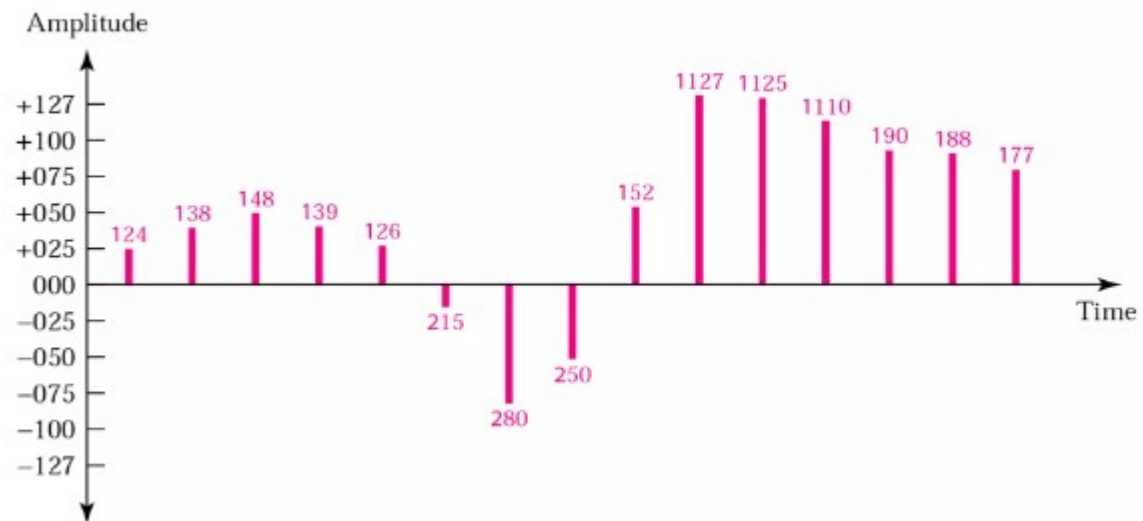
- Is the process of obtaining amplitudes of a signal at a regular interval
 - The analog signal is sampled every T interval.
- Most important factor in sampling is the rate at which analog signal is sampled.
- According to Nyquist Theorem, the sampling rate must be at least two times of the highest frequency of the signal.



Quantization

- Sampling yields discrete form of continuous analog signal. Every discrete pattern shows the amplitude of the analog signal at that instance.
- The quantization is done between the maximum amplitude value and the minimum amplitude value.
- Quantization is approximation of the instantaneous analog value.

Quantized PAM signal

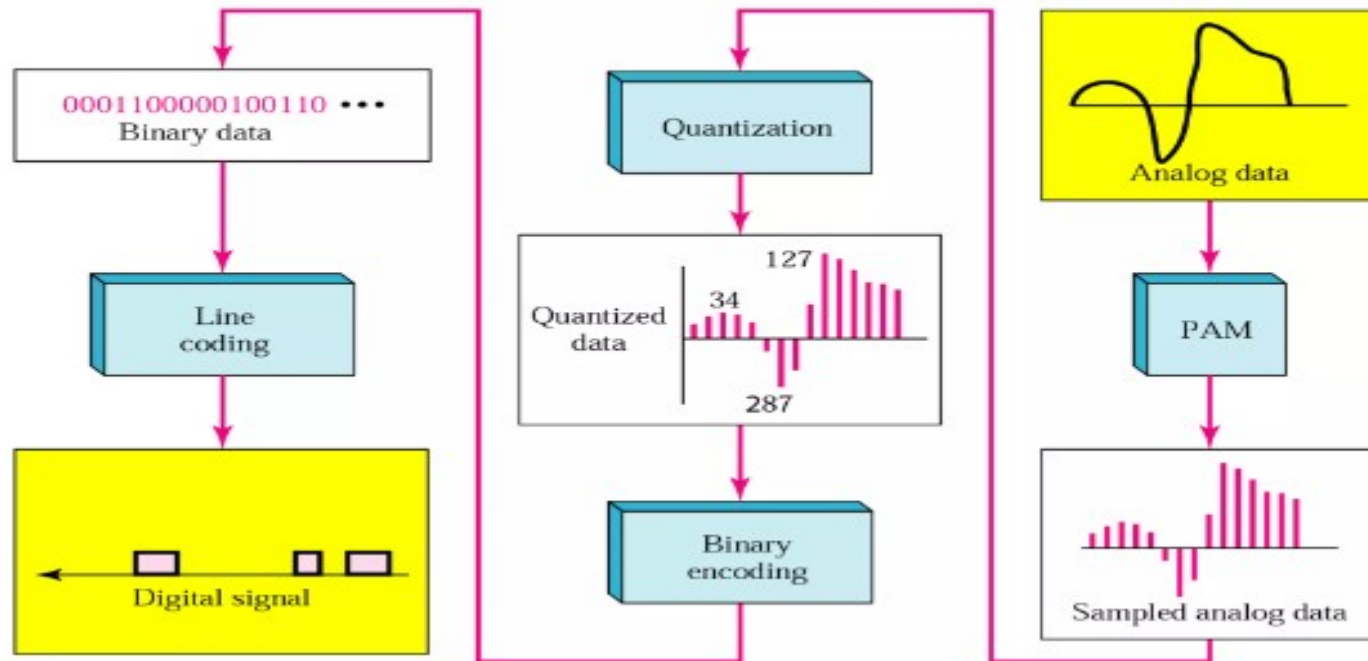


Quantizing by using sign and magnitude

+024	00011000	-015	10001111	+125	01111101
+038	00100110	-080	11010000	+110	01101110
+048	00110000	-050	10110010	+090	01011010
+039	00100111	+052	00110110	+088	01011000
+026	00011010	+127	01111111	+077	01001101

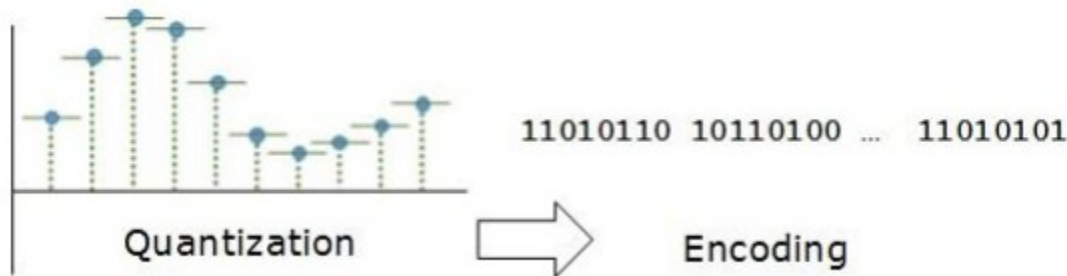
Sign bit
+ is 0 - is 1

Digitization of an Analog Signal



Encoding

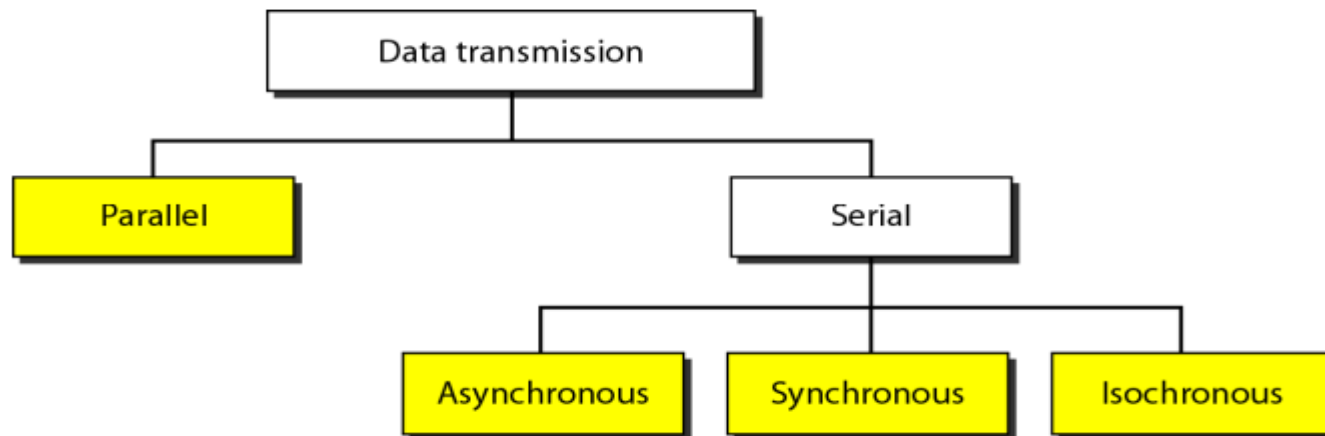
- In encoding, each approximated value is then converted into binary format.



Transmission Modes

- The transmission mode decides how data is transmitted between two computers.
- The binary data in the form of 1s and 0s can be sent in two different modes: **Parallel** and **Serial**.

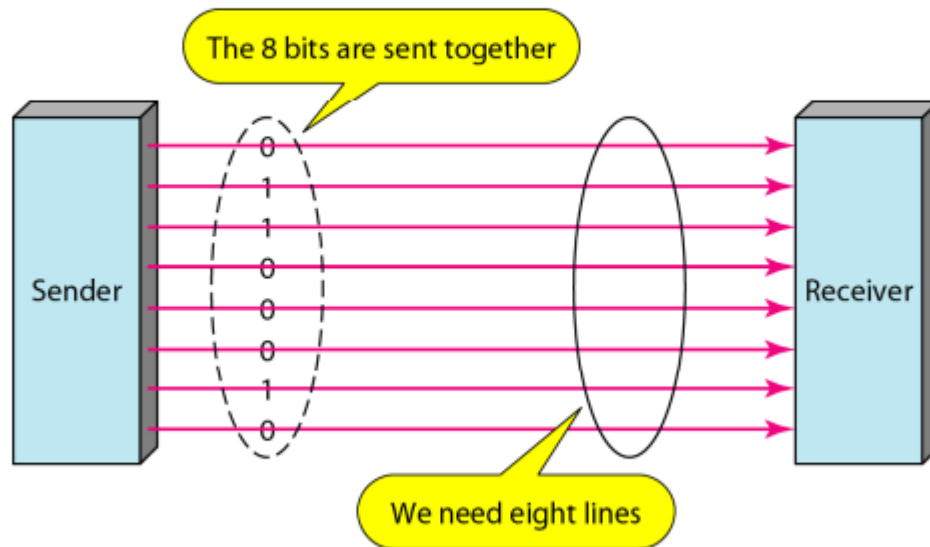
Data transmission and modes



Parallel Transmission

- Bits in a group are sent simultaneously, each using a separate link
- n wires are used to send n bits at one time
- Advantage : speed
- Disadvantage: cost; limited to short distance

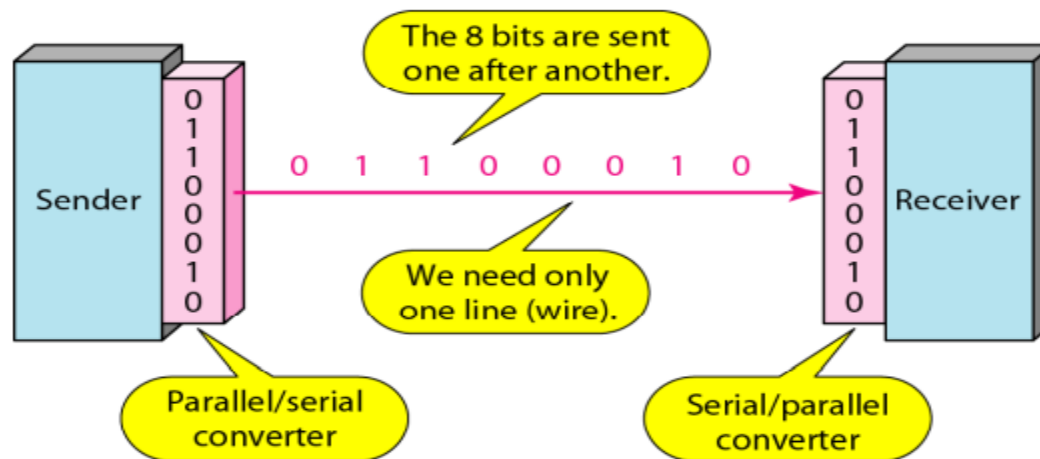
Parallel transmission



Serial Transmission

- Transmission of data one bit at a time using only one single link
- **Advantage:** reduced cost
- **Disadvantage:** requires conversion device
- Methods
 - **Asynchronous**
 - **synchronous.**

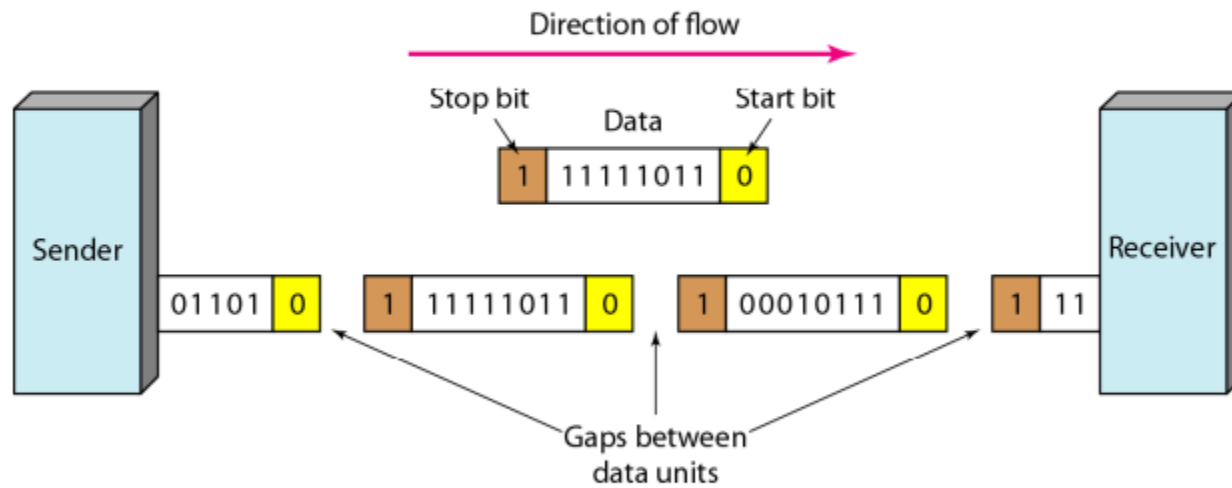
Serial transmission



Asynchronous Serial Transmission

- It is named so because there is timing is not important.
- Transfer of data with start bit and stop bits and a variable time interval(gap) between data unites
- Start bit alerts receiver that new group of data is arriving
- Stop bits alert receiver that byte is finished
- Synchronization achieved through start/stop bits with each byte received

Asynchronous transmission



Note

In asynchronous transmission, we send 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte. There may be a gap between each byte.

Note

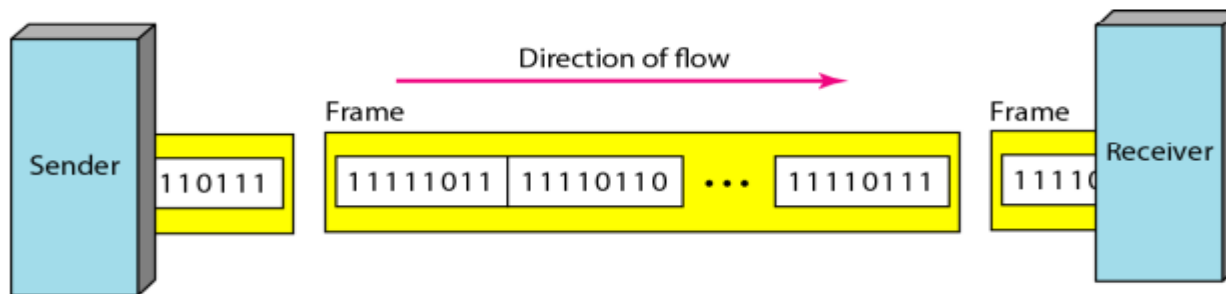
**Asynchronous here means
“asynchronous at the byte level,”
but the bits are still synchronized;
their durations are the same.**

- Requires additional overhead(stop/start bits)
- Slower
- Cheap and effective
- Ideal for low-speed communication when gaps may occur during transmission(ex. Keyboard)

Synchronous Serial Transmission

- Data bits are sent in burst mode without maintaining gap between bytes (8- bits). Single burst of data bits may contain a number of bytes.
- **Timing** in synchronous transmission has importance as there is no mechanism followed to recognize start and end data bits.
- It is up to the receiver to recognize and separate bits into bytes.
- **Advantage:** high speed, and it has no overhead of extra header and footer bits as in asynchronous transmission.
- Byte synchronization is accomplished by data link layer

Synchronous transmission



Note

In synchronous transmission, we send bits one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits.

Isochronous

- A sequence of events is isochronous if the events occur regularly, or at equal time intervals. The isochronous transmission guarantees that the data arrive at a fixed rate.
- In real-time audio and video, in which uneven delays between frames are not acceptable, synchronous transmission fails. For example, TV images are broadcast at the rate of 30 images per second; they must be viewed at the same rate.

- If each image is sent by using one or more frames, there should be no delays between frames. For this type of application, synchronization between characters is not enough; the entire stream of bits must be synchronized. The isochronous transmission guarantees that the data arrive at a fixed rate.

- Reference

- Data communications and networking 5th edition
by Behrouz a. Fourouzan